

WT Unique Questions

Module I: Fundamentals of Wireless Communication

#	Question	Marks	Month & Year
1	Explain Direct Sequence Spread Spectrum.	5	Dec 2023
2	Compare FDMA, TDMA and CDMA.	5	Dec 2023
3	How does OFDMA handle synchronization and interference between users?	5	Dec 2024
4	What is Frequency Hopping Spread Spectrum (FHSS), and how does it improve communication security?	5	Dec 2024
5	Explain how LoRaWAN enables long-range, low-power communication by using spread spectrum techniques (Chirp Spread Spectrum).	5	Dec 2024
6	State the advantages of TDMA over FDMA.	5	Dec 2025
7	Differentiate between FHSS and DSSS.	5	Dec 2025
8	Describe evolution of 1G to 5G mobile systems.	5	May 2023
9	Explain the spread spectrum and briefly outline DSSS and FHSS.	5	May 2024
10	What do you mean by massive MIMO; Compare 1G - 5G mobile standards.	10	Dec 2023
11	What do you mean by massive MIMO, Compare 1G - 5G mobile standards.	5	May 2025
12	Compare and contrast 1G to 5G based on the technological differences and advancements. Write a short note on Massive MIMO.	10	May 2024
13	Compare Frequency Hopping spread spectrum and Direct Sequence Spread Spectrum.	5	May 2025
14	Draw and explain OFDMA technique for multiplexing.	10	Dec 2025
15	Draw and explain OFDMA with a neat diagram.	10	May 2024

Module II: Wide Area Wireless Networks

#	Question	Marks	Month & Year
1	Explain in detail GSM Architecture with neat diagram.	10	Dec 2023
2	List and explain the core components of the UMTS network?	5	Dec 2024
3	Explain the function of a Subscriber Identity Module (SIM) in GSM.	5	Dec 2024
4	Write note on UMTS security.	5	Dec 2025
5	Write short note on: EDGE for 2.5G GSM and IS-136.	10	Dec 2025
6	Draw and explain the GSM time slot hierarchy.	10	Dec 2025
7	Write note on LTE frame structure in detail.	5	May 2023
8	Write note on CDMA2000.	5	May 2023
9	Draw and explain UMTS network architecture and compare GSM and UMTS.	10	May 2023
10	Draw neat diagram of GPRS system architecture and explain the function of each block in it.	10	May 2023
11	Illustrate and explain GSM architecture with its interfaces. Compare GSM and UMTS.	10	May 2024
12	Explain role of SGSN and GGSN in GPRS.	5	May 2025
13	What are the different types of GSM logical Channels? Explain in detail.	10	May 2025
14	Explain UMTS and GSM security.	10	Dec 2023
15	Explain UMTS and GSM security.	10	May 2024
16	Draw and explain LTE network architecture in detail.	10	Dec 2025
17	Draw and explain 4G network architecture with its specifications.	10	Dec 2023
18	Explain the architecture and working of 4G.	10	Dec 2024
19	Explain 4G network architecture with its specifications.	10	Dec 2025
20	Draw and explain 4G network architecture and compare with 3G.	10	May 2025

Module III: Wireless Metropolitan and Local Area Networks

#	Question	Marks	Month & Year
1	State the features of Wi-MAX.	5	Dec 2023

#	Question	Marks	Month & Year
2	Draw and explain IEEE 802.11 Architecture.	10	Dec 2023
3	How does Wi-Fi handle multiple users and avoid collisions in the network?	10	Dec 2024
4	Explain the architecture and Features of a WiMAX network.	10	Dec 2024
5	Compare IEEE802.11 and IEEE802.16.	5	May 2025
6	Draw and explain protocol architecture of IEEE802.11. Also discuss on power management in IEEE802.11 infrastructure network.	10	May 2023
7	Draw and Explain Wi-Fi Architecture and Protocol Architecture.	10	May 2024
8	State the Features of Wi-MAX. Draw and explain the architecture of Wi-MAX.	10	May 2024
9	Describe the system architecture of 802.11.	10	May 2025

Module IV: Wireless Personal Area Networks and Ad hoc Networks

#	Question	Marks	Month & Year
1	Explain the difference between an Infrastructures based Network and an Adhoc Network of WLAN.	5	Dec 2023
2	What are the various challenges in WSN and explain the architecture of WSN.	10	Dec 2023
3	Explain Zigbee protocol stack.	10	Dec 2023
4	Give the Features of VANET, E-VANET and MANET.	5	Dec 2023
5	Explain difference between Adhoc network and infrastructure based network.	5	Dec 2024
6	What role does VANET play in intelligent transportation systems (ITS)?	5	Dec 2024
7	What is mesh networking in ZigBee, and how does it enhance the network's reliability?	10	Dec 2024
8	Explain the different features of VANET and E-VANET.	5	Dec 2025
9	Draw and explain MANET. What are the design challenges in MANET?	10	Dec 2025

#	Question	Marks	Month & Year
10	Draw and explain wireless sensor network communication architecture.	10	Dec 2025
11	Explain any four features of MANET and compare MANET and WSN.	5	May 2023
12	What is Zigbee. Explain ZigBee protocol stack in detail.	10	May 2023
13	Describe Bluetooth architecture and protocol Stack. Also discuss its limitations.	10	May 2023
14	Explain the layered architecture of WSN protocol and discuss issues and challenges in WSN.	10	May 2023
15	Give the Features of VANET and MANET.	5	May 2024
16	Compare Infrastructures based Network and an Adhoc Network.	5	May 2024
17	Outline the WSN architecture along with neat diagrams. State the various issues in WSN.	10	May 2024
18	State different features of Zigbee and explain its protocol stack.	10	May 2024
19	Explain the difference between an Infrastructures based Network and an Adhoc Network of WLAN.	5	May 2025
20	What are the various challenges in WSN and explain the architecture of WSN.	10	May 2025
21	Explain Zigbee protocol stack.	10	May 2025
22	Compare VANET and MANET. State the features of LoRaWAN.	10	May 2025
23	Draw and explain Bluetooth Architecture.	10	May 2025

Module V: Wireless Network Security

#	Question	Marks	Month & Year
1	Give the significance of WEP protocols. What are the features of WPA2.	10	Dec 2023
2	Explain the pairing and security mechanisms used in Bluetooth communication.	10	Dec 2024
3	What are the primary security mechanisms employed in GSM to protect user data and privacy?	10	Dec 2024

#	Question	Marks	Month & Year
4	Discuss the flaws in WEP's initialization vector (IV) implementation and how it leads to key reuse and susceptibility to attacks.	10	Dec 2024
5	Give the significance of WEP protocols. What are the features of WPA2.	10	May 2023
6	Explain WEP and the security standard used.	5	May 2024
7	Give the significance of WEP protocols. What are the features of WPA2.	10	May 2025
8	With the help of neat diagram, explain the digital signature process.	10	Dec 2025

Module VI: Wireless Network Design Considerations

#	Question	Marks	Month & Year
1	Outline the method that supports mobility in CISCO Unified Wireless Network.	5	Dec 2023
2	Draw and explain Cisco UWN lightweight AP and WLC Operation.	10	Dec 2023
3	Outline the method that supports mobility in CISCO Unified Wireless Network.	5	Dec 2024
4	How does Cisco implement security in its Unified Wireless Network?	10	Dec 2024
5	Outline the method that supports mobility in CISCO Unified Wireless Network.	5	Dec 2025
6	Draw and explain the architecture of Cisco UWN with its features.	10	Dec 2025
7	Outline the method that supports mobility in CISCO Unified Wireless Network.	5	May 2023
8	Draw and explain the architecture of Cisco UWN with its features.	10	May 2023
9	Illustrate the RF site survey process and their importance in the design process of designing a wireless network with Lightweight AP and WLC.	10	May 2024
10	Outline the method that supports mobility in CISCO Unified Wireless Network.	5	May 2024

#	Question	Marks	Month & Year
11	Outline the method that supports mobility in CISCO Unified Wireless Network.	5	May 2025
12	Give the features of Cisco Unified Wireless Network and explain its architecture.	10	May 2025